Acoustic systems（4th class）

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- Wave equation in three dimensional system
- Sound pressure level
- Point source
- Directivity

Sound propagation in three dimension

Operation of the sound in the three dimensions

In the three dimensions, the particle velocity has to be treated as vector.

$$\boldsymbol{v} = (v_x , v_y , v_z)$$

Also, partial derive of the spatial variable has to be operated in three dimensions as following

$$\frac{\partial}{\partial x} \quad \Rightarrow \quad \nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

Wave equation in the three dimensions

Equation of motion : $$\rho_0 \frac{\partial \mathbf{v}}{\partial t} = -\nabla p = \left(-\frac{\partial p}{\partial x}, -\frac{\partial p}{\partial y}, -\frac{\partial p}{\partial z} \right)$$

Conservation of mass: $$\frac{\partial \rho}{\partial t} = -\rho_0 \nabla \bullet \mathbf{v} = -\rho_0 \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

Gas state equation: $$p = c^2 \rho \quad (where \quad c^2 = \frac{\gamma P_0}{\rho_0})$$

Wave equation in the three dimensions

$$\nabla^2 p = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2}$$

Helmholtz equation

In the acoustic phenomena, the time variation is expressed by the harmonic function $e^{j\omega t}$

Sound pressure

$$p = \rho \frac{\partial \Phi}{\partial t} = j\omega\rho\Phi$$

Particle velocity

$$\mathbf{v} = -\nabla\Phi$$

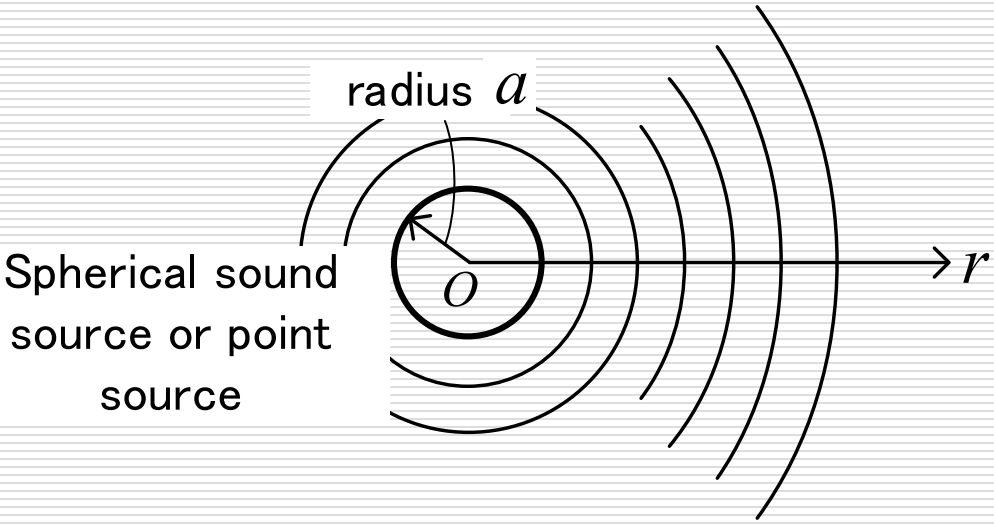
By substituting above equation to the wave equation, the following equation "Helmholtz equation" is obtained.

$$\nabla^2\Phi + k^2\Phi = 0$$

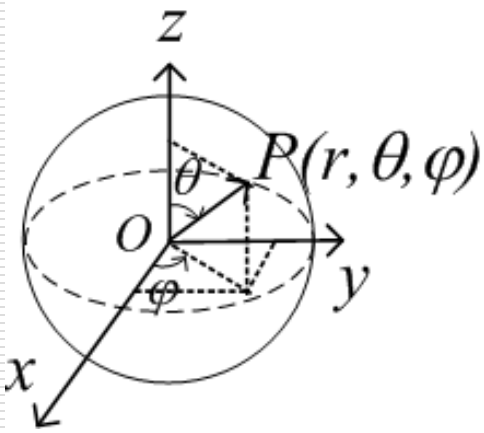
Polar coordinate system and point source

Propagation wave emitted from spherical sound source or point source is spherical wave which propagates uniformly in all directions.

Sound pressure is not distributed on the circumferential direction (θ, φ).



Polar coordinate system



Spherical wave

Wave equation in the polar coordinate system is following.

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \Phi}{\partial r} \right) + k^2 \Phi = 0$$

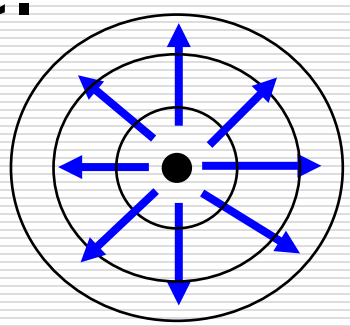
Solution of the above equation is following.

$$\Phi(x,t) = \left(\frac{C_1}{r} e^{-jkr} + \frac{C_2}{r} e^{jkr} \right) e^{j\omega t}$$

Froward and backward spherical wave

Forward wave is expansion into sound field from point source.

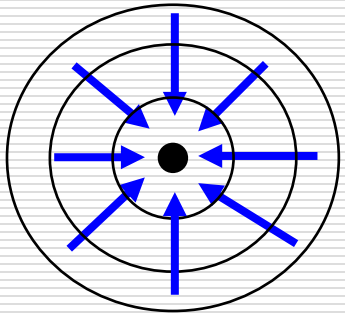
$$\Phi(x,t) = \frac{C_1}{r} e^{j(\omega t - kr)}$$



We only have to consider forward wave in 3 dimensional propagation.

Backward wave is contraction into point source. Generally, it's hard to propagate.

$$\Phi(x,t) = \frac{C_2}{r} e^{j(\omega t + kr)}$$



Sound wave from point source: sound pressure p

In the case of no reflection that is, only forward wave exists, velocity potential is as following,

$$\Phi(r,t) = \frac{C_1}{r} e^{j(\omega t - kr)}$$

Generally, only forward wave propagates from the spherical point source.

Sound pressure p can be can be obtained by the time derivative of the velocity potential as following,

$$p(r,t) = \rho \frac{\partial \Phi}{\partial t} = j\omega\rho\Phi$$

Hereafter, static density is expressed by ρ instated of ρ_0.

Sound wave from point source: particle velocity v

Particle velocity v can be obtained by the spatial derivative of the velocity potential as following,

$$v = -\frac{\partial \Phi}{\partial r} = -\frac{\partial}{\partial r}\left(\frac{C_1}{r} e^{j(\omega t - jkr)}\right) = -\frac{\partial}{\partial r}\left(\frac{C_1}{r}\right) e^{j(\omega t - jkr)} - \frac{C_1}{r}\frac{\partial\left(e^{j(\omega t - jkr)}\right)}{\partial r}$$

$$= -\left(-\frac{C_1}{r^2} e^{j(\omega t - jkr)}\right) - \frac{C_1}{r}(-jkr) e^{j(\omega t - jkr)} = \frac{C_1}{r^2}(1 + jkr) e^{j(\omega t - jkr)}$$

$$= \frac{1 + jkr}{r}\Phi$$

Sound pressure level

Sense of hearing

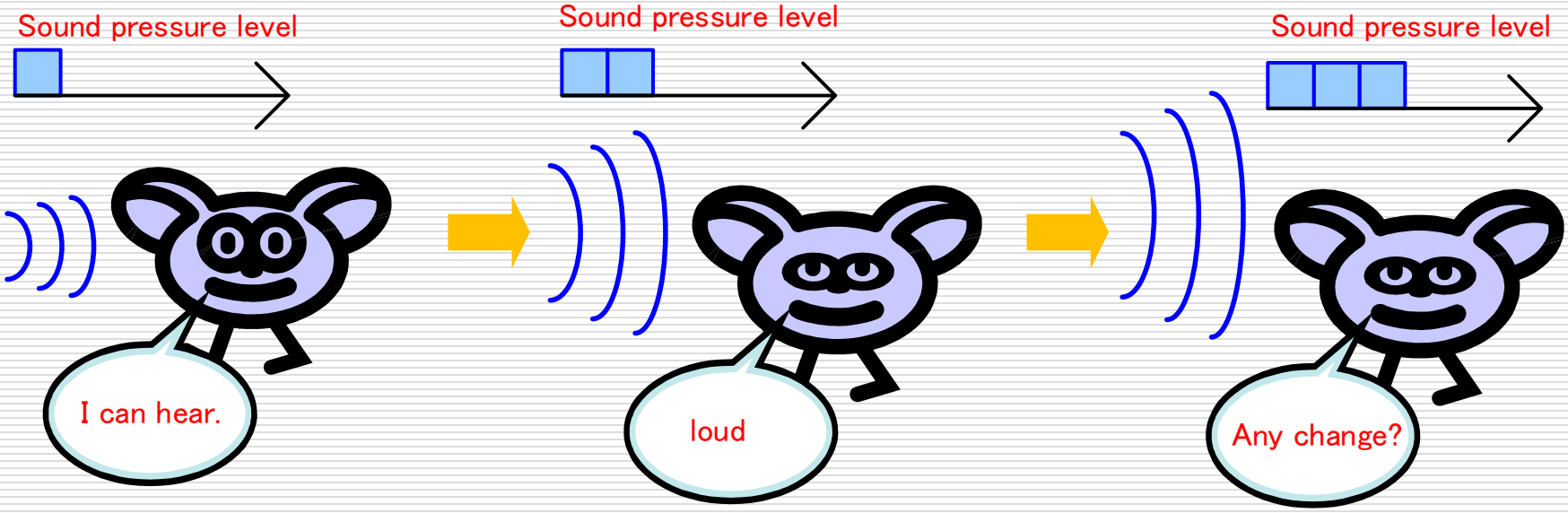
Unit of sound pressure is [Pa] which is same as static pressure.

The human sense amount of hearing is not always proportional to sound pressure level.

The human feels loudness comparing to the original sound level.

Sense of hearing about sound level(1)

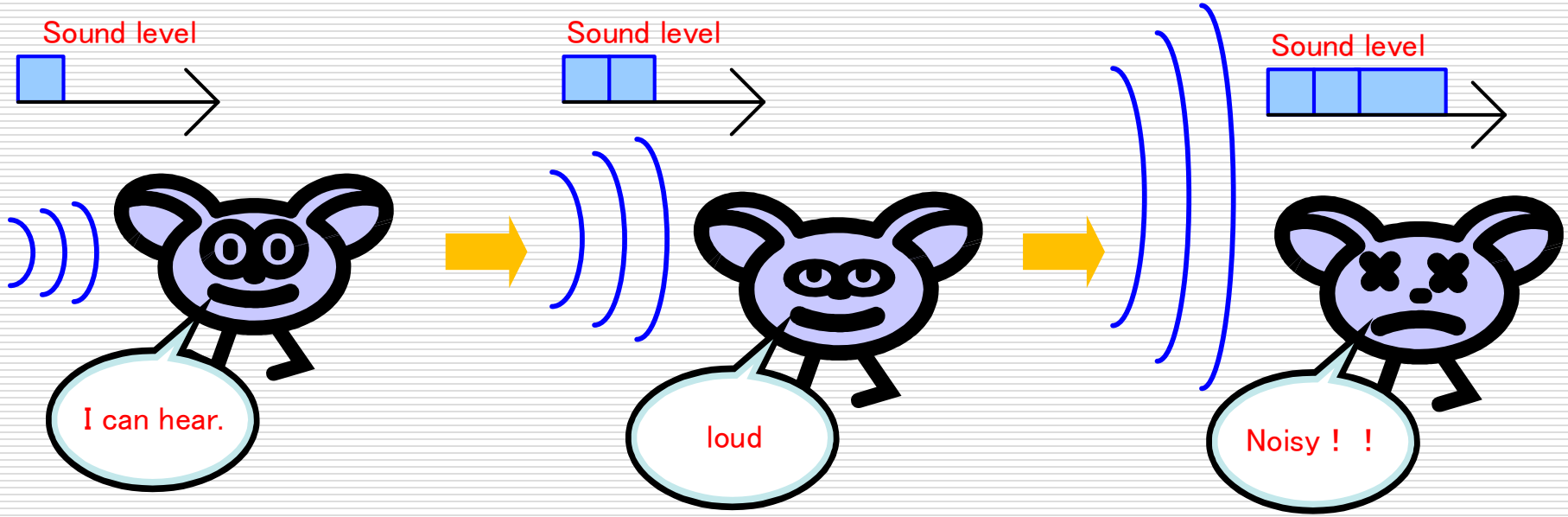
In the case that sound pressure level increases gradually at the same magnitude.



We <u>can't</u> feel louder at the large sound level.

Sense of hearing about sound level

In the case that sound pressure level increases in the proportion to the original sound, that is logarithmically ...



We can feel louder at the large sound level.

Sound pressure level

It is appropriate to treat sound pressure level logarithmically.

Sound pressure is expressed by the following.
Sound Pressure Level [S.P.L.]

$$\text{S.P.L.} = 20\, log_{10}\left(\frac{p}{p_{ref}}\right)\ [\text{dB}]$$

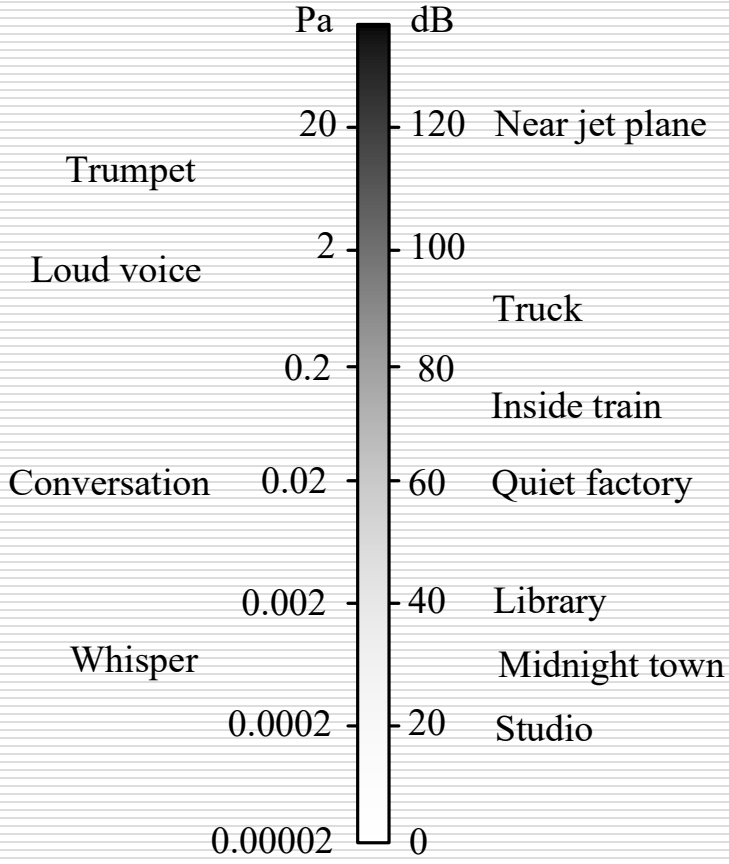
where reference is $p_{ref} = 20\ [\mu Pa]$

p_{ref} corresponds to the minimum sound pressure which the people can hear.

Example of S.P.L. in the life

Example S.P.L. against several situations.

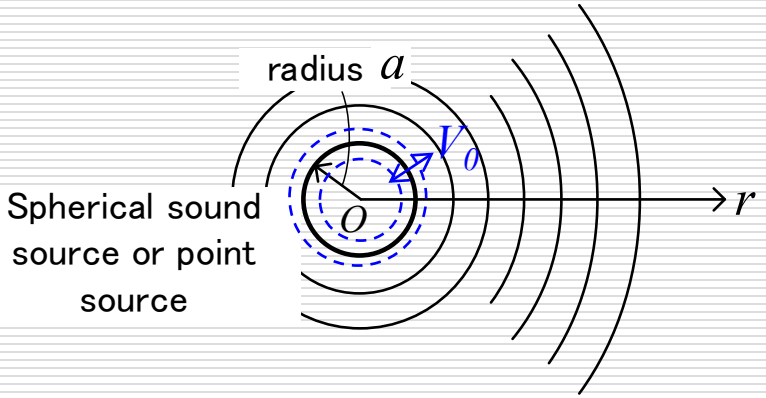
Please calculate the particle velocity for each S.P.L.



Volume velocity

Particle velocity V_0 at the surface of the spherical sound source is as follows,

$$V_0 = -\left.\frac{\partial \Phi}{\partial r}\right|_{r=a} = \frac{C_1}{a^2}\left(1 + jka\right)e^{j(\omega t - ka)}$$



When the sound source almost can be treated as a point source, that is $a << \lambda$, particle velocity is re-written as $V_0 = C_1/a^2$, and velocity potential is as follows

$$\Phi = \frac{V_0 a^2}{r} e^{j(\omega t - kr)} = \frac{4\pi a^2 V_0}{4\pi r} e^{j(\omega t - kr)}$$

$Q = 4\pi a^2 V_0$ is called as "volume velocity".

Sound field emitted by point source

Velocity potential and sound pressure are re-written as follows,

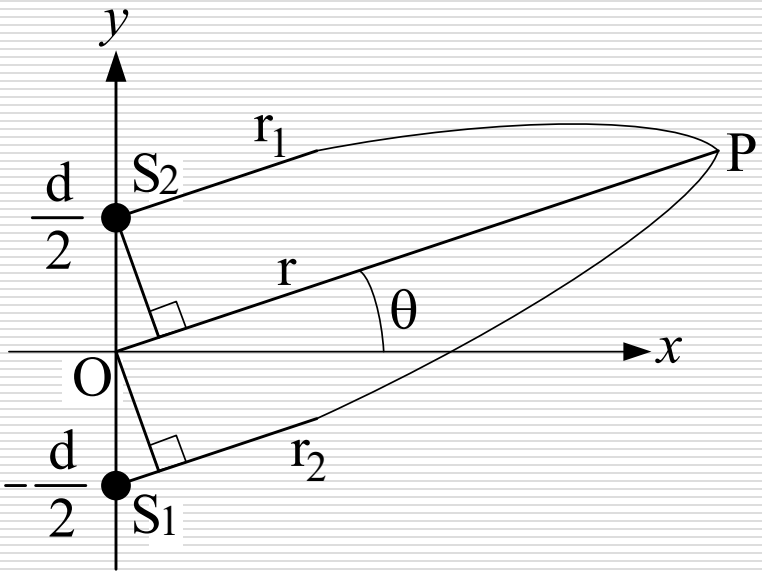
$$\Phi = \frac{Q}{4\pi r} e^{j(\omega t - kr)} \quad , \qquad p = \rho \frac{\partial \Phi}{\partial t} = j\omega\rho \frac{Q}{4\pi r} e^{j(\omega t - kr)}$$

Point source creates diffusion sound field where sound pressure becomes a tenth when the distance from the source becomes ten times.

Sound source like point source is called as non-directional sound source, or as omni-directional sound source.

Sound field emitted by two point sources

When two separate point sources are emitted in phase as shown in the right, the velocity potential at P is given by followings.



$$\Phi = \left(\frac{Q}{4\pi r_1} e^{-jkr_1} + \frac{Q}{4\pi r_2} e^{-jkr_2} \right) e^{j\omega t}$$

Sound field emitted by two point sources at far field.

In the far field from the source, r >> d, $r_1 \approx r_2 \approx r$ and the velocity potential is re-rewritten as follows.

$$\Phi = \left(\frac{Q}{4\pi r} e^{-jk\left(r + \frac{kd}{2}\sin\theta\right)} + \frac{Q}{4\pi r} e^{-jk\left(r - \frac{kd}{2}\sin\theta\right)} \right) e^{j\omega t}$$

$$\Phi = \frac{Q}{2\pi r} e^{j(\omega t - kr)} \cos\left(\frac{kd}{2}\sin\theta\right)$$

Sound field formed by two point sources

Please derive the following eq.(1) from eq.(2).

eq.(1) $$\Phi = \frac{Q}{2\pi r} e^{j(\omega t - k r)} \cos\left(\frac{k d}{2}\sin\theta\right)$$

eq.(2)

$$\Phi = \left(\frac{Q}{4\pi r} e^{-jk\left(r + \frac{k d}{2}\sin\theta\right)} + \frac{Q}{4\pi r} e^{-jk\left(r - \frac{k d}{2}\sin\theta\right)} \right) e^{j\omega t}$$

Directivity index

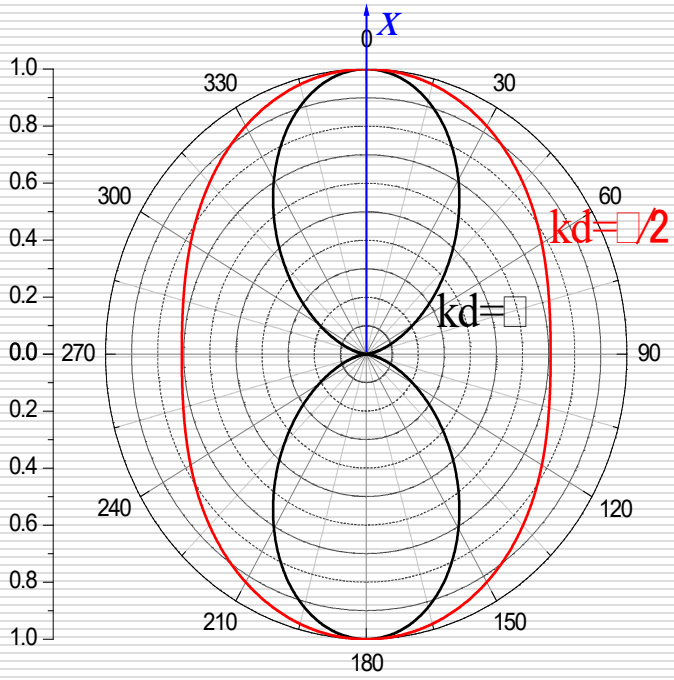
Directivity index $D(\theta)$ written by following indicates the sound pressure distribution in the same distance from sound source as a function of direction angle θ.

$$D(\theta) = \left| \cos\left(\frac{k\,d}{2} \sin\theta \right) \right|$$

where maximum sound pressure is 1 by normalization.

Directional pattern

Right figure shows the sound pressure distribution along direction angle θ at $kd=\pi$ and $kd=\pi/2$.

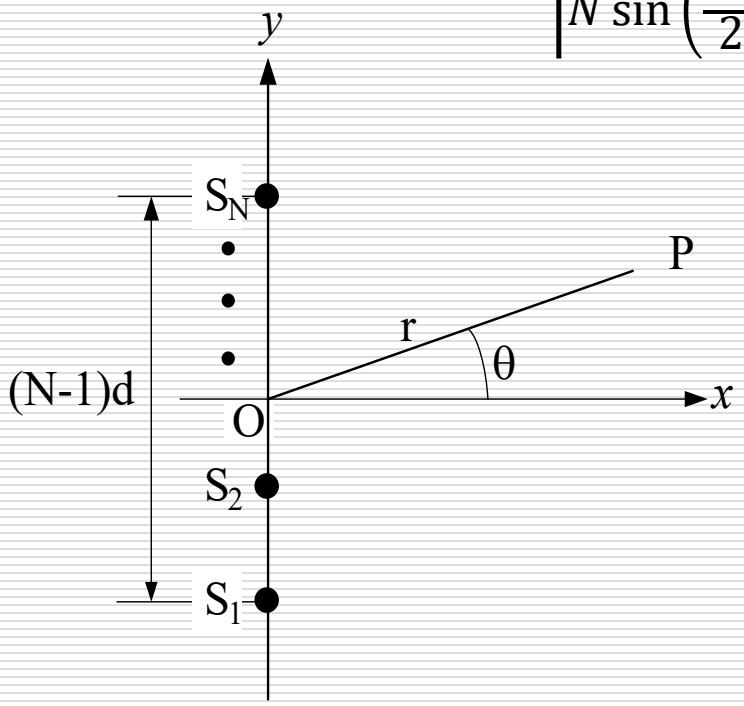


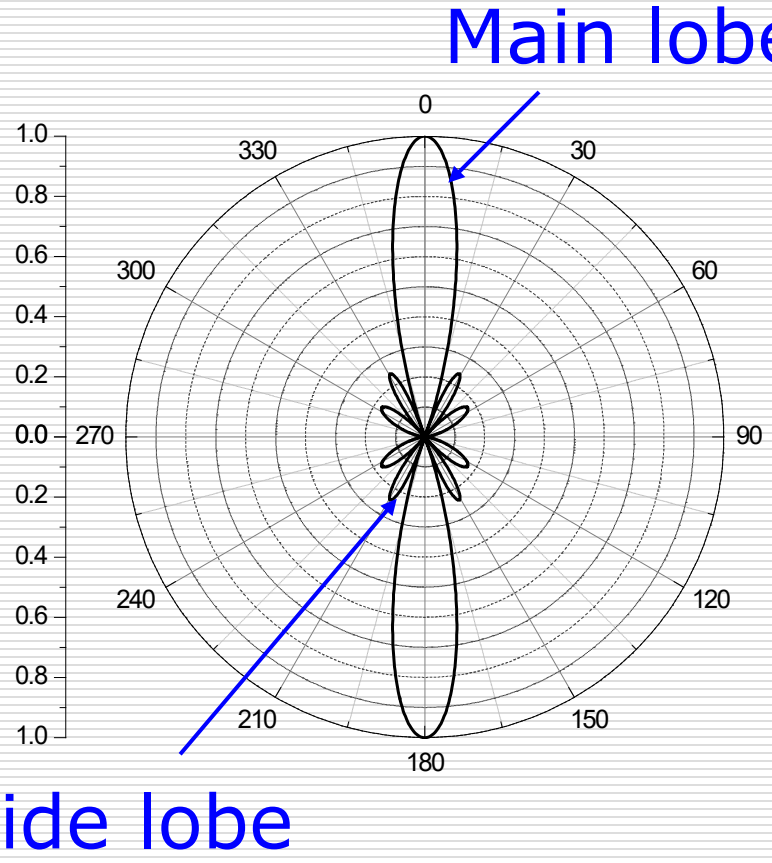
Right figure is called as "directional pattern".

Directional pattern of multiple point sources

Directivity index of multiple point sources is given by

$$D(\theta) = \left| \frac{\sin\left(\frac{Nkd}{2}\sin\theta\right)}{N\sin\left(\frac{kd}{2}\sin\theta\right)} \right|$$





Question 1

Draw the directivity patterns on the graph respectively when the distance of the two sound and frequency are following. In drawing, the sound velocity is 340m/s.

(a) Frequency f : 680Hz
Source distance d : 25cm

(b) Frequency f : 340Hz
Source distance d : 12.5cm

